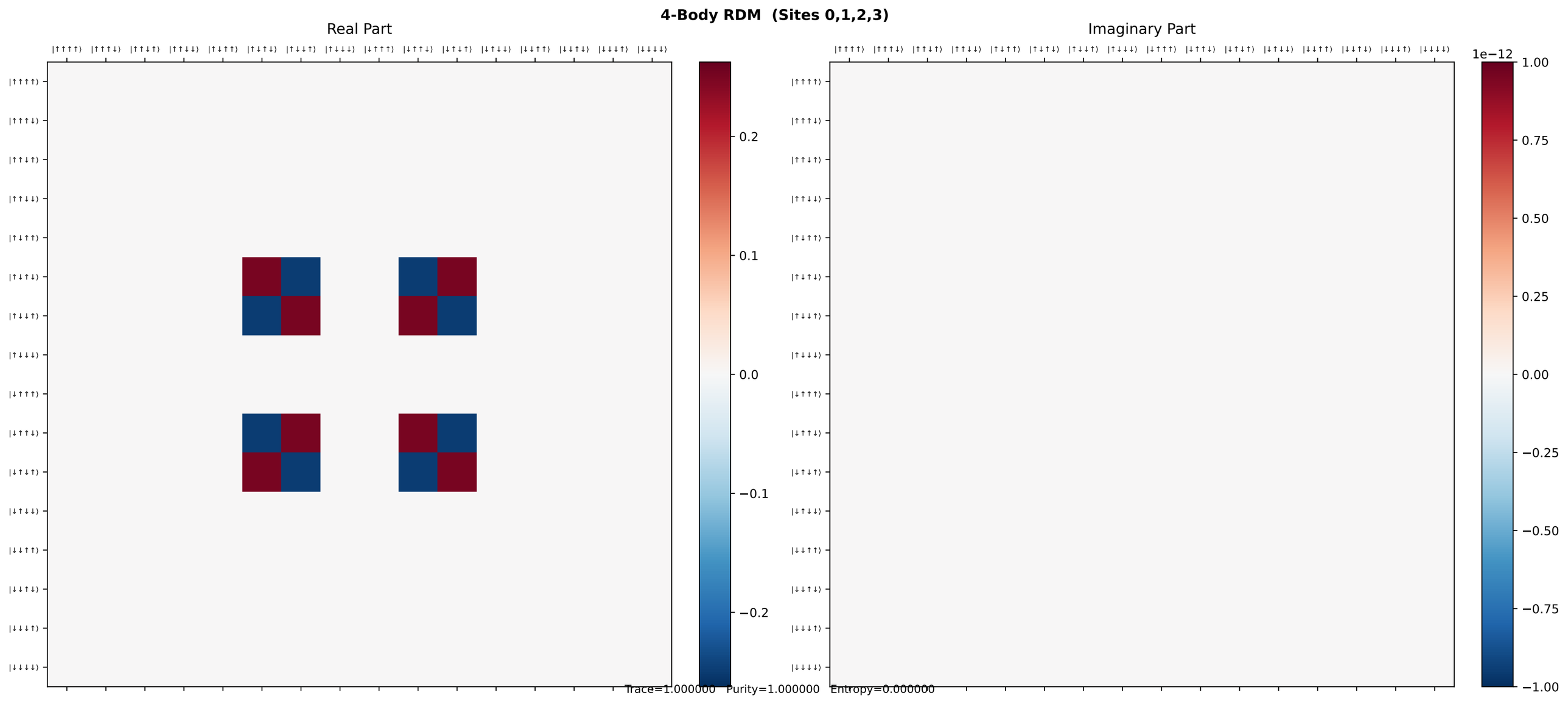
Genuine Multipartite Entanglement Analysis

```
Model                              Ground-state properties           GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E₀                 : -2.47213595  Extracted n-spin RDM : n = 4
                                   E₁                 : -1.23606798
System parameters                                                    Purity:
-----------------                  ⟨S²⟩               : 0.000000      γ = Tr(ρ²)
Sites (L)          : 4             S                  : 0.0000
Electrons (Ne)     : 4             ⟨Sz⟩               : 0.000000    Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1  : 1.0           γ_A = 1  ↔  ρ = ρ_A ⊗ ρ_B
Interaction (U)    : 2.0           Spin state         : Singlet (S=0)
Mode               : dimerized                                     C_GME (Troiani et al.):
n-body order       : 4             Double occupancy                  C_GME = √(2 · min_A (1 − Tr(ρ_A²)))
                                   ----------------
                                   ⟨D⟩ total          : 0.552786    Decision logic:
                                   ⟨D⟩ / site         : 0.138197      γ = 1  and  all Tr(ρ_A²) < 1   → GME
                                                                      γ = 1  and  some Tr(ρ_A²) = 1  → No GME
                                                                      γ < 1                          → Mixed (convex-roof needed)

                                                                    Date: 2026-03-04  14:58
```


4-Body RDM (Sites 0,1,2,3)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=1.000000 Entropy=0.000000

```
RDM Metrics  (Sites = 0,1,2,3)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 1.0000000000
  Von Neumann entropy    : 0.0000000000
  Eigenvalue  min        : -6.964014e-17
  Eigenvalue  max        : 1.000000e+00
```

GME Analysis — 4-Body RDM (Sites 0,1,2,3)

Bipartitions (Sites 0,1,2,3)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 1.00000000$

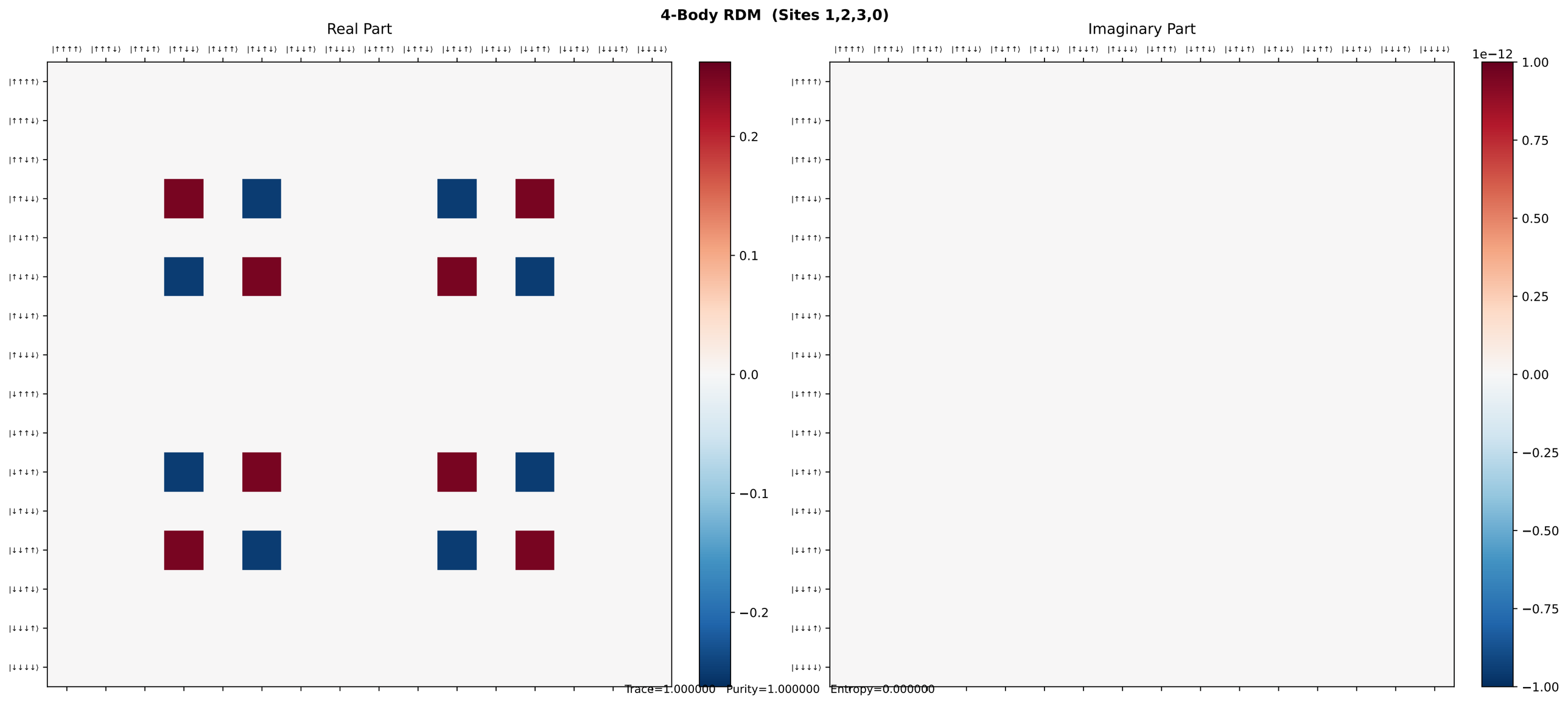
A	B	$\mathrm{Tr}(\rho_A^2)$	
(0,)	(1, 2, 3)	0.50000000	
(0, 1)	(2, 3)	1.00000000	←
(0, 2)	(1, 3)	0.25000000	
(0, 3)	(1, 2)	0.25000000	
(0, 1, 2)	(3,)	0.50000000	
(0, 1, 3)	(2,)	0.50000000	
(0, 2, 3)	(1,)	0.50000000	

STATUS: NO GME

At least one bipartition satisfies $\mathrm{Tr}(\rho_A^2) = 1 \rightarrow \rho = \rho_A \otimes \rho_B$

Separable cuts:

A=(0, 1) | B=(2, 3)


4-Body RDM (Sites 1,2,3,0)
Real Part
Imaginary Part
1e−12
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000

```
RDM Metrics  (Sites = 1,2,3,0)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 1.0000000000
  Von Neumann entropy    : 0.0000000000
  Eigenvalue  min        : -5.251745e-17
  Eigenvalue  max        : 1.000000e+00
```

GME Analysis — 4-Body RDM (Sites 1,2,3,0)

Bipartitions (Sites 1,2,3,0)

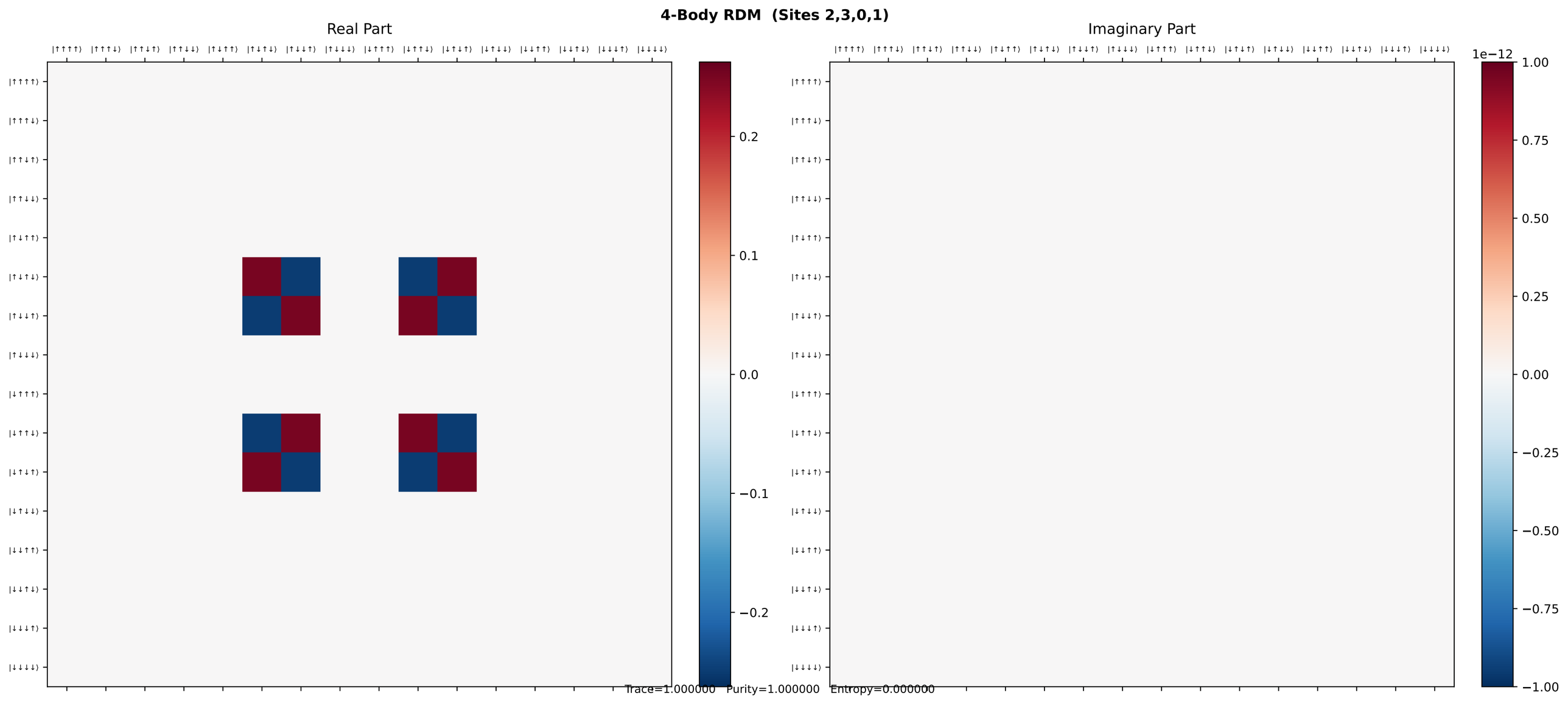
Global purity γ = Tr(ρ²) = 1.00000000

A	B	Tr(ρ_A²)	
(1,)	(2, 3, 0)	0.50000000	
(1, 2)	(3, 0)	0.25000000	
(1, 3)	(2, 0)	0.25000000	
(1, 0)	(2, 3)	1.00000000	←
(1, 2, 3)	(0,)	0.50000000	
(1, 2, 0)	(3,)	0.50000000	
(1, 3, 0)	(2,)	0.50000000	

STATUS: NO GME

At least one bipartition satisfies Tr(ρ_A²) = 1 → ρ = ρ_A ⊗ ρ_B

Separable cuts:
  A=(1, 0) | B=(2, 3)


4-Body RDM (Sites 2,3,0,1)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=1.000000 Entropy=0.000000

```
RDM Metrics  (Sites = 2,3,0,1)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 1.0000000000
  Von Neumann entropy    : 0.0000000000
  Eigenvalue  min        : -6.938894e-17
  Eigenvalue  max        : 1.000000e+00
```

GME Analysis — 4-Body RDM (Sites 2,3,0,1)

Bipartitions (Sites 2,3,0,1)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 1.00000000$

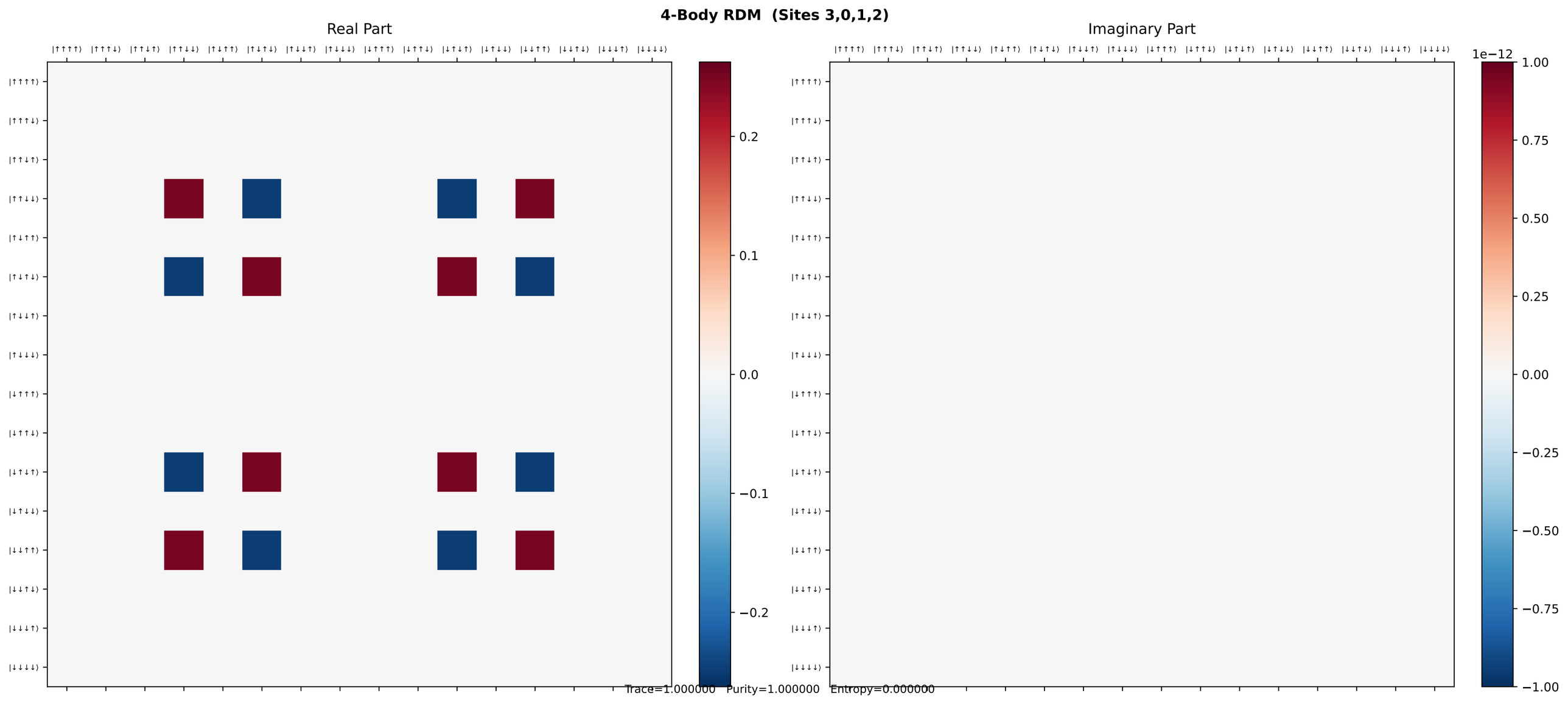
A	B	$\mathrm{Tr}(\rho_A^2)$	
(2,)	(3, 0, 1)	0.50000000	
(2, 3)	(0, 1)	1.00000000	←
(2, 0)	(3, 1)	0.25000000	
(2, 1)	(3, 0)	0.25000000	
(2, 3, 0)	(1,)	0.50000000	
(2, 3, 1)	(0,)	0.50000000	
(2, 0, 1)	(3,)	0.50000000	

STATUS: NO GME

At least one bipartition satisfies $\mathrm{Tr}(\rho_A^2) = 1 \rightarrow \rho = \rho_A \otimes \rho_B$

Separable cuts:

A=(2, 3) | B=(0, 1)


4-Body RDM (Sites 3,0,1,2)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=1.000000 Entropy=0.000000

```
RDM Metrics  (Sites = 3,0,1,2)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 1.0000000000
  Von Neumann entropy    : 0.0000000000
  Eigenvalue  min        : -7.343435e-17
  Eigenvalue  max        : 1.000000e+00
```

GME Analysis — 4-Body RDM (Sites 3,0,1,2)

Bipartitions (Sites 3,0,1,2)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 1.00000000$

A	B	$\mathrm{Tr}(\rho_A^2)$	
(3,)	(0, 1, 2)	0.50000000	
(3, 0)	(1, 2)	0.25000000	
(3, 1)	(0, 2)	0.25000000	
(3, 2)	(0, 1)	1.00000000	←
(3, 0, 1)	(2,)	0.50000000	
(3, 0, 2)	(1,)	0.50000000	
(3, 1, 2)	(0,)	0.50000000	

STATUS: NO GME

At least one bipartition satisfies $\mathrm{Tr}(\rho_A^2) = 1 \rightarrow \rho = \rho_A \otimes \rho_B$

Separable cuts:
A=(3, 2) | B=(0, 1)